

```

WARNING: Using incubator modules: jdk.incubator.vector
100% 5183/5183 [00:00<00:00, 89006.67it/s]
qrels key sample: '1' (type: str)
query_list key sample: '1' (type: str)
Apr 04, 2026 10:37:50 PM org.apache.lucene.store.MemorySegmentIndexInputProvider <init>
INFO: Using MemorySegmentIndexInput with Java 21; to disable start with -Dorg.apache.lucene.store.MMapDirectory.enableMemorySegments=false
BM25 Search: 100% 300/300 [00:03<00:00, 88.77it/s]

=====
Run: BM25 Baseline
nDCG@10 : 0.6816
Recall@100 : 0.9280
judged@10 : 0.0883
=====
RM3 Search: 100% 300/300 [00:07<00:00, 38.83it/s]

=====
Run: BM25 + RM3
nDCG@10 : 0.6576
Recall@100 : 0.9203
judged@10 : 0.0900
=====
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

2026-04-04 22:38:02,395 - huggingface_hub.utils._http - WARNING - Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable h
Loading weights: 100% 199/199 [00:00<00:00, 1263.29it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: castorini/tct_colbert-v2-hnp-msmarco
Key | Status | |
-----+-----+
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Dense Encoding & Search: 100% 300/300 [00:37<00:00, 7.92it/s]

=====
Run: Dense (Zero-shot)
nDCG@10 : 0.5751
Recall@100 : 0.8643
judged@10 : 0.0797
=====

=====
Run: RRF k=60 (BM25 + Dense)
nDCG@10 : 0.6690
Recall@100 : 0.9437
judged@10 : 0.0883
=====

=== FINAL RESULTS SUMMARY ===
Run nDCG@10 Recall@100 judged@10
-----
BM25 0.6816 0.9280 0.0883
BM25+RM3 0.6576 0.9203 0.0900
Dense 0.5751 0.8643 0.0797
RRF (BM25+Dense) 0.6690 0.9437 0.0883

Results saved to results.json

Launch interactive query mode? [y/N]: y

=====
INTERACTIVE QUERY MODE (BM25)
Type a query to retrieve passages. Leave blank to exit.
=====

Query> smoking causes lung cancer

Top 5 results for: 'smoking causes lung cancer'
-----
[1] Score: 9.6819 | ID: 7639744
Title: Influence of smoking cessation after diagnosis of early stage lung cancer on prognosis: systematic review of observational studies with meta-analysis
Snippet: OBJECTIVE To systematically review the evidence that smoking cessation after diagnosis of a primary lung tumour affects prognosis. DESIGN Systematic re

[2] Score: 9.0487 | ID: 26067999
Title: Screening for Lung Cancer: U.S. Preventive Services Task Force Recommendation Statement
Snippet: The U.S. Preventive Services Task Force (USPSTF) makes recommendations about the effectiveness of specific preventive care services for patients withou

[3] Score: 8.1904 | ID: 16422880

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[3] Score: 8.1904 | ID: 16422880
    Title: Effect of smoking reduction on lung cancer risk.
    Snippet: CONTEXT Many smokers are unable or unwilling to completely quit smoking. A proposed means of harm reduction is to reduce the number of cigarettes smoke

[4] Score: 8.1521 | ID: 25725663
    Title: Wedelolactone protects human bronchial epithelial cell injury against cigarette smoke extract-induced oxidant stress and inflammation responses through
    Snippet: Cigarette smoke is the leading cause of the development of various lung diseases including lung cancer through triggering oxidant stress and inflammato

[5] Score: 7.6701 | ID: 21495419
    Title: The impact of COPD on lung health worldwide: epidemiology and incidence.
    Snippet: Information on the prevalence of COPD was obtained from vital statistics, health interview surveys, hospital charge records, national publications, and
-----

Query> aspirin reduces heart attack risk

Top 5 results for: 'aspirin reduces heart attack risk'
-----

[1] Score: 8.1213 | ID: 15648443
    Title: Long-term effect of aspirin on cancer risk in carriers of hereditary colorectal cancer: an analysis from the CAPP2 randomised controlled trial
    Snippet: BACKGROUND Observational studies report reduced colorectal cancer in regular aspirin consumers. Randomised controlled trials have shown reduced risk of

[2] Score: 7.9123 | ID: 7613033
    Title: Aspirin in the chemoprevention of colorectal neoplasia: an overview.
    Snippet: Considerable evidence supports the effectiveness of aspirin for chemoprevention of colorectal cancer (CRC) in addition to its well-established benefits

[3] Score: 7.3120 | ID: 7662206
    Title: A Century of Cholesterol and Coronaries: From Plaques to Genes to Statins
    Snippet: One-fourth of all deaths in industrialized countries result from coronary heart disease. A century of research has revealed the essential causative age

[4] Score: 6.9743 | ID: 33118292
    Title: Optimal therapy for reduction of lipoprotein(a).
    Snippet: WHAT IS KNOWN AND OBJECTIVE There is a growing body of experimental and clinical evidence for the atherogenic and pro-thrombotic potential of Lipoprote

[5] Score: 6.9459 | ID: 43334921
    Title: Association of aspirin and NSAID use with risk of colorectal cancer according to genetic variants.
    Snippet: IMPORTANCE Use of aspirin and other nonsteroidal anti-inflammatory drugs (NSAIDs) is associated with lower risk of colorectal cancer. OBJECTIVE To iden
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Query>
```